Dorsaf Sallami

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Research Interests

My research focuses on responsible AI for information integrity, with an emphasis on automated fact-checking and fake news detection. I study how language models and human–AI collaboration can support verification, and how to evaluate these systems beyond accuracy, including fairness, explainability, contextual ambiguity, and societal impact.

- Member, Centre for the Study of Democratic Citizenship.

Academic Positions

Postdoctoral Researcher, Mila – Québec AI Institute Jan 2026 – Present

- Advisors: Reihaneh Rabbany (McGill University) and Jean-François Godbout (Université de Montréal).
- Project lead for *Veracity*, an open-source AI fact-checking initiative:
  - Cited in Canada's National Artificial Intelligence Strategy as an example of impactful Canadian AI research.
  - Added to the Senate of Canada's resources for senators and staff in May 2026.

Education

Ph.D. **Université de Montréal**, Computer Science — Sept 2022 – Dec 2025

- **Supervisors:** Professor Esma Aïmeur and Professor Gilles Brassard
- Thesis: Toward socially responsible artificial intelligence approaches for fake news detection

M.Sc. **Université de Montréal**, Computer Science — Sept 2020 – Aug 2022

- **Supervisor:** Professor Esma Aïmeur
- Thesis: Personalized fake news aware recommendation system

Eng. **Higher Institute of Applied Sciences and Technology, Tunisia** — Sept 2017 – Aug 2020

- Engineering degree in Software Engineering
- Advanced systems design and software engineering

Prep. **Higher Institute of Applied Sciences and Technology, Tunisia** — Sept 2015 – Aug 2017

- Preparatory cycle for the national engineering program
- Mathematics, Physics, and Computer Science

Professional Development

Stanford University, *Ethics, Technology, and Public Policy for Practitioners* Fall 2026

- Live cohort-based professional program on technology ethics, public policy, and responsible innovation.

Selected Publications

Journal Articles

1. SALLAMI, Dorsaf, and AÏMEUR, Esma. **Explain Further: Multi-Level Explanations for Fake News Detection Using Large Language Models**. *International Journal of Information Technology*, 2025.
2. SALLAMI, Dorsaf, and AÏMEUR, Esma. **Exploring Beyond Detection: A Review on Fake News Prevention and Mitigation Techniques**. *Journal of Computational Social Science*, vol. 8, article 23, 2025.
3. SALLAMI, Dorsaf, and AÏMEUR, Esma. **Fairframe: A Fairness Framework for Bias Detection and Mitigation in News**. *AI and Ethics*, vol. 5, pp. 2467–2483, 2025.

4. ABDALLAH, Wejden, KANZARI, Dalel, SALLAMI, Dorsaf, et al. **A Deep Reinforcement Learning Based Decision-Making Approach for Avoiding Crowd Situation Within the Case of Covid'19 Pandemic**. *Computational Intelligence*, vol. 38, no. 2, pp. 416–437, 2022.

Conference Papers

5. SALLAMI, Dorsaf, and AÏMEUR, Esma. **Aletheia: Detect, Discuss, and Stay Informed on Fake News**. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence (IJCAI)*, Demo Track, pp. 11109–11113, 2025.
6. AÏMEUR, Esma, BRASSARD, Gilles, and SALLAMI, Dorsaf. **Too Focused on Accuracy to Notice the Fallout: Towards Socially Responsible Fake News Detection**. *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society (AIES)*, vol. 8, no. 1, pp. 55–65, 2025.
7. SALLAMI, Dorsaf, and AÏMEUR, Esma. **Does Gender Matter? Examining and Mitigating Gender Bias in Fake News Detection**. In *Foundations and Practice of Security (FPS 2024)*, Springer, pp. 249–266, 2025.
8. SALLAMI, Dorsaf, BEN SALEM, Rim, and AÏMEUR, Esma. **Trust-Based Recommender System for Fake News Mitigation**. In *Adjunct Proceedings of the 31st ACM Conference on User Modeling, Adaptation and Personalization (UMAP)*, pp. 104–109, 2023.
9. AMRI, Sabrine, SALLAMI, Dorsaf, and AÏMEUR, Esma. **EXMULF: An Explainable Multimodal Content-Based Fake News Detection System**. In *Foundations and Practice of Security (FPS 2021)*, Springer, pp. 177–187, 2022.

Workshop Papers

10. SALLAMI, Dorsaf, GUEDDICHE, Ahmed, and AÏMEUR, Esma. **From Hype to Reality: Revealing the Accuracy and Robustness of Transformer-Based Models for Fake News Detection**. In *3rd Workshop on Adverse Impacts and Collateral Effects of Artificial Intelligence Technologies (AiOfAi)*, IJCAI, 2023.

Honors and Scholarships

- **2025–2026:** Rector's Honour List, Université de Montréal.
- **2020–2025:** Tunisian Government Scholarship for studies in Canada — ranked first nationally in the 2020 selection, Tunisian Ministry of Higher Education.
- **Fall & Winter 2022, 2023, 2024, 2025:** DIRO Excellence-Writing-Support Scholarship, Université de Montréal

Teaching and Supervision

Courses and Tutoring

Teaching Assistant, Université de Montréal

- **IFT2015/IFT6002 – Data Structures** *Winter 2023, 2024, 2025*
  Led laboratory sessions, assessed assignments, and provided individualized support during office hours.
- **IFT6261 – Knowledge Processing** *Winter 2024*
  Assessed assignments and supported students during office hours.

Tutor, Tutoring 2.0 Program, Université de Montréal *Fall 2022*

- Provided academic support to undergraduate students in computer science courses.

Invited Lectures

- **Guest Lecturer**, Knowledge Processing (IFT 6261), Université de Montréal: "Information Integrity in the Age of Generative AI" — October 2026 | Montréal, Canada
- **Guest Lecturer**, Knowledge Processing (IFT 6261), Université de Montréal: "Multimodal Content-Based Fake News Detection System" — February 2022 | Montreal, Canada

Doctoral Research Mentorship

Sukanya Krishna, Ph.D. Student — **Harvard University**, Jan. 2026 – Present

- Research project: *Ambiguity and Underspecification in AI-Assisted Fact-Checking*

Undergraduate Research Supervision and Mentorship

Tong Wu, Undergraduate Research Intern — **McGill University**, Summer 2026

- Research project: *Multi-Turn Fact-Checking with Large Language Models*

Andrew Lin, Undergraduate Research Intern — **McGill University**, Summer 2026

- Research project: *Claim Augmentation for Fact-Checking*

Hugh Kelly, Undergraduate Research Intern — **McGill University**, Winter 2026

- Research project: *Source Preferences Affect Trust in Large Language Models*

Arpad Botond Rigo, Undergraduate Research Intern — **Université de Montréal**, Summer 2023

- Research project: *Simulation of Fake News Propagation*

Academic Service and Leadership

Editorial Service

- **Topic Editor** — *"AiOfAi: Adverse Impacts and Collateral Effects of Artificial Intelligence Technologies"*, *Frontiers in Artificial Intelligence*, 2026

Workshop and Event Organization

- **Bridge Program Organiser** — Bridging AI and Democracy: Towards Trustworthy Information Ecosystems, AAAI 2027.
- **Workshop Organiser** — Disinformation and Misinformation under the lens of Generative AI Safety, AAAI 2027.
- **Workshop Organiser** — 5th Workshop on Adverse Impacts and Collateral Effects of Artificial Intelligence Technologies, The ACM Web Conference 2027
- **Social Event Organiser** — *Who Decides What Counts as True?*, ACM FAccT 2026, June 2026, Montréal; sponsored by Mastercard.
- **Workshop Organiser**— 4th Workshop on Adverse Impacts and Collateral Effects of Artificial Intelligence Technologies, The ACM Web Conference 2026 —Proceedings
- **Publicity Chair** — 3rd Workshop on Adverse Impacts and Collateral Effects of Artificial Intelligence Technologies, 32nd International Joint Conference on Artificial Intelligence (IJCAI), 2023 —Proceedings

Research Coordination and Expert Contributions

- **Designated Coordinator** — IVADO Thematic Semester: Information Integrity in the Age of Generative AI, 2027
- **Contributing Expert** — Global Trust Challenge, 2026

Reviewing

- **Reviewer for** — International Joint Conference on Artificial Intelligence (IJCAI); AAAI/ACM Conference on AI, Ethics, and Society (AIES); Conference on Neural Information Processing Systems (NeurIPS); AAAI; and AI and Ethics Journal

Media Coverage and Public Engagement

- **Radio Interview** — "L'État de New York veut empêcher les jeunes de confondre l'IA avec des humains", *Y a pas deux matins pareils*, Radio-Canada Toronto — November 9, 2026 | Interview in French
- **Co-Author, Opinion Article** — "Construire la confiance, au-delà de l'urgence d'adopter l'intelligence artificielle", *Le Devoir*, July 2026
- **Media Interview**, Knowable Magazine article, "AI shows promise in the fight against fake news", interviewed by journalist Katya Zimmer — June 2026
- **Research Media Coverage**, UdeMNouvelles article "Fake news: do the tools meant to protect us actually mask their flaws?" featuring my research on socially responsible fake news detection; also covered by Tech Xplore and shared by Crickit.ai — March 2026
- **Content Contributor** — CYBER project (HEC Montréal, IMC2, Centre de formation en langues des affaires), "Comment éviter la propagation de fausses informations en ligne ?" — 2025

Talks and Presentations

Invited Talks and Seminars

- **Invited Speaker**, LATECE Seminar, Université du Québec à Montréal (UQAM): "Peut-on confier la vérification des faits à l'IA ? Promesses, limites et perspectives" — October 2026 | Montréal, Canada | Presented in French
- **Invited Speaker**, Community of Inquiry in AI Ethics, Curtin University: "Why Accuracy Isn't Enough When AI Decides What Counts as True" — July 2026 | Online
- **Invited Speaker**, First Canada–Denmark Municipal Knowledge Exchange Webinar: "RAG4MuniESG: ESG Assessment for Municipal Sustainability through Retrieval-Augmented Generation, a Quebec Case Study" — June 2025 | Online

Conference and Public Presentations

- **Poster Presenter**, 8th AAAI/ACM Conference on AI, Ethics, and Society (AIES): "Too Focused on Accuracy to Notice the Fallout: Towards Socially Responsible Fake News Detection" — October 2025 | Madrid, Spain
- **Presenter**, Forum INCYBER Canada (delegation of the Centre for Cybersecurity Belgium) — October 2025 | Montreal, Canada
- **Poster Presenter**, 34th International Joint Conference on Artificial Intelligence (IJCAI): "Aletheia: Detect, Discuss, and Stay Informed on Fake News" — August 2025 | Montreal, Canada
- **Poster Presenter**, Journée de la Recherche du DIRO: "CoALFake: Collaborative Active Learning with Human-LLM Co-Annotation for Cross-Domain Fake News Detection" — May 2025 | Montreal, Canada
- **Presenter**, 17th International Symposium on Foundations & Practice of Security (FPS): "Does Gender Matter? Examining and Mitigating Gender Bias in Fake News Detection" — December 2024 | Montreal, Canada
- **Presenter**, Innovate4Cities Conference: "What Do Policy, Planning, and Reporting Documents Reveal About Urban Sustainability Initiatives? Patterns and Insights from AI-Assisted Textual Analysis" — September 2024 | Montreal, Canada
- **Presenter**, 15th International Sustainability Transitions Conference (IST): "Nature and Biodiversity in Urban Sustainability Transition Planning and Reporting Documents: Patterns and Insights from AI-Assisted Textual Analysis" — June 2024 | Oslo, Norway (Online)
- **Presenter**, Cyberconference: "Beyond the Facts: Fake News, Generative Artificial Intelligence, and Cybersecurity" — April 2024 | Montreal, Canada
- **Presenter**, Inaugural Event of the Multidisciplinary Institute for Cybersecurity and Cyber Resilience (IMC^2): "Explainable System for Countering Fake News and Doxxing" — October 2023 | Montreal, Canada

- **Presenter**, 3rd Workshop on Adverse Impacts and Collateral Effects of Artificial Intelligence Technologies: "From Hype to Reality: Transformer-Based Models for Fake News Detection Performance and Robustness Revealed" — August 2023 | Macao, China (Online)
- **Poster Presenter**, 31st ACM Conference on User Modeling, Adaptation and Personalization (UMAP) — June 2023 | Limassol, Cyprus
- **Presenter**, Annual Canadian Celebration of Women in Computing Conference (CanCWIC): "Multimodal Content-Based Fake News Detection System" — October 2022 | Toronto, Canada

Research and Industry Experience

Circle Medical, AI Research Intern — Montreal, June 2025 – Sep 2025
- Designed and implemented an evaluation framework for ICD-10/CPT code prediction and HPI/Plan summary generation
- Developed quantitative metrics and procedures to measure model performance systematically.
- Benchmarked and compared several LLMs (OpenAI, Anthropic).

AlphaFixe Capital (Mitacs), AI Research Intern (Part-time) — Montreal, Jan 2024 – May 2025
- Research intern in Artificial Intelligence applied to sustainable finance
- Python, NLP, Data Analysis, Data collection, LLM

Videotron, Data Analyst Intern — Montreal, May 2023 – Aug 2023
- Data Cleaning, Data Integration, Data Visualization
- Python, NetSuite, Databases

Société Générale, BI - Data Analyst Intern (Part-time) — Montreal, Sep 2021 – April 2022
- Data Cleaning, Data Integration, Data Visualization
- Python , JS , GitHub , JSON , Databases, REST/POST Data visualisation, Power BI

ITPEAC, End of Studies Internship — Tunisia, Feb 2020 – July 2020
- An approach to optimizing decision-making in a crisis based on deep reinforcement learning. Case Study: COVID-19

Extra-Curricular Activities

- **2021–2025:** Lab Leader, *AI Lab for Cybersecurity*, Montreal
- **2019–2020:** Training Manager, *ARSII Association*, Tunisia
- **2018–2020:** Treasurer, *ISSATSo Data Science Club*, Tunisia
- **Personal Interest:** When I'm not debugging code or mentoring students, you'll probably find me hiking up a trail

Technical Skills

Programming Languages: Python, Java **Web Development:** Angular.js, Flask **Machine Learning:** PyTorch, NLTK, Scikit-learn, Keras, OpenCV **Documentation:** LaTeX, Markdown

Languages

Arabic (Native), French (Professional), English (Professional)